

Trust Calibration in Human–Agent Sequential Graph Colouring: Scenario, Scoring, and Agent Design

1 Scenario

An undirected graph $G = (V, E)$ is partitioned among three participants — one human h and two AI agents a_1, a_2 — with disjoint node clusters $V = V_h \sqcup V_{a_1} \sqcup V_{a_2}$. Each participant colours their own nodes from $\mathcal{C} = \{\text{red, blue, green}\}$ *one at a time* in a fixed sequential order: all V_h nodes first, then V_{a_1} , then V_{a_2} . No node may be skipped or reordered.

All colourings are valid — there are no hard constraints. Performance is measured entirely through a *scoring function* (Section 2). Adjacent same-colour pairs incur a penalty. The human always acts first, so their choices structurally constrain the agents’ downstream options.

Sessions consist of up to $A = 3$ attempts on the same graph. Between attempts the participant sees their score and may start again from scratch.

Optional random order. A configurable flag shuffles the human-node sequence (and optionally interleaves agent nodes) each attempt, seeded deterministically by $s + k$ where s is the session seed and k the attempt number. The planning agent (Mode 2) re-plans against the actual shuffled order.

2 Scoring Model

Every colour decision earns points for *all* participants simultaneously. Table ?? shows the default per-node values.

Table 1: Per-node points per participant. Each row is a colour; shared total is the column sum. Bold = each participant’s individually preferred colour.

Colour	Human	a_1	a_2	Shared
Red	4	1	2	7
Blue	1	4	2	7
Green	2	2	4	8

Each participant’s preferred colour differs; green is globally shared-optimal (8 pts). Choosing red (human preference) sacrifices 1 shared point per node relative to green, creating a built-in individual–team tension.

Clash penalty. For edge $(u, v) \in E$ with $x_u = x_v$, a penalty of $\delta = 10$ is deducted. If $\text{owner}(u) = \text{owner}(v)$, the full δ falls on that participant. Otherwise, $\lfloor \delta/2 \rfloor + (\delta \bmod 2)$ is deducted from the owner of u and $\lfloor \delta/2 \rfloor$ from the owner of v . For $\delta = 10$ this is a 5/5 split. The shared score is $\sum_p \text{Score}_p$. The theoretical maximum is $8|V|$ (all green, no clashes).

3 Experimental Conditions

Three within-subjects conditions differ in how agent advice is delivered.

Mode 1 — Manual Collaborative. Before each human-owned node is coloured, both agents simultaneously issue a colour proposal with a written rationale and a score-prediction table. The human reads these and chooses freely. Agent-owned nodes are coloured automatically by the agents’ greedy rule (Section 4.1).

Mode 2A — Autonomous, High Quality ($q = 0.8$). Before the session starts, the planning agent generates a complete proposed colouring (Section 4.2). This plan is shown one step at a time; the human may accept or override each human-node step. Agent steps are auto-applied.

Mode 2B — Autonomous, Low Quality ($q = 0.3$).

Interface and interaction are *identical* to Mode 2A. Only the quality parameter differs. Participants are not informed of the difference.

Comparing 2A vs. 2B with identical presentation is the core manipulation: does behavioural trust calibration track underlying quality or interface appearance?

4 Agent Algorithms

All agent logic is deterministic and rule-based. No learned model or LLM is used.

4.1 Proposal Agent (Mode 1)

Runs independently for each agent at each human-owned node v .

1. Find $\text{adj} = \{u \in V_a \mid (u, v) \in E\}$. If empty, skip.
2. Choose reference node: prefer already-coloured adjacent nodes; else first alphabetically.
3. If reference is *coloured*: recommend first colour not clashing with its assigned colour. If reference is *uncoloured*: recommend first colour that is not the agent’s own preferred colour (anticipating placing preferred colour there later).
4. Fall back to $\mathcal{C}[0]$ if all colours excluded.

Short rationale template:

- Clash case: “My node X is [col] — recommend [col] to avoid a clash.”
- Future case: “I plan to colour X [preferred] — recommend [col] to coordinate.”

A *score-prediction table* is also shown: for each colour the human could choose, the agent forward-simulates its remaining greedy decisions and reports its expected total.

Auto-colouring (agent-owned nodes): find adjacent colours already placed; among clash-free candidates, pick the colour maximising the agent’s own per-node points.

4.2 Planning Agent (Mode 2)

Generates a complete plan before colouring begins.

Algorithm 1 `PlanningAgent.generate_plan( $k, \sigma, q, s$ )`

Require: attempt k , node sequence σ , quality q , seed s

```

1:  $\text{rng} \leftarrow \text{Random}(s + k)$ ;  $A \leftarrow \{\}$ 
2: for all  $v \in \sigma$  do
3:    $\text{clash} \leftarrow \{A[u] \mid (u, v) \in E, u \in A\}$ 
4:    $C \leftarrow \mathcal{C} \setminus \text{clash}$ ; if  $C = \emptyset$  then  $C \leftarrow \mathcal{C}$ 
5:    $c^* \leftarrow \arg \max_{c \in C} \text{shared}(c)$ 
6:   if  $\text{rng.random}() < q$  then
7:      $\text{chosen} \leftarrow c^*$  // shared-optimal
8:   else
9:      $\text{chosen} \leftarrow \text{rng.choice}(C)$  // random safe
10:  end if
11:   $A[v] \leftarrow \text{chosen}$ 
12: end for
13: return plan steps with rationales

```

The sub-agents vote self-interestedly (each votes for its own preferred colour) before the selection step; votes are recorded in the *deliberation log* shown to participants on request, but do not influence the outcome. Rationale templates differ by selection method:

- Shared-optimal: “[Colour] gives the team N pts, maximising shared score.”
- Random: “[Colour] selected after deliberation (team score: N pts).”

5 Interface Summary

The Tkinter window (1020×660 px min) has three zones: (i) a **scoring HUD** (shared score prominently; per-participant cards); (ii) a **main pane** with an overview graph canvas (440×280 px), a neighbourhood detail canvas (440×210 px), and a mode-specific right panel; (iii) a **sequence bar** showing colouring progress (one cell per node, grey until coloured).

Mode 1 right panel. Reward key $\rightarrow$ active node label $\rightarrow$ score-prediction grid (agents $\times$ colours, $\star$ on each agent’s preferred) $\rightarrow$ colour-choice buttons (enabled after proposals shown).

Mode 2 right panel. Proposed colour + short rationale $\rightarrow$ ACCEPT / EXPLAIN / MODIFY buttons $\rightarrow$ collapsible full rationale (Text widget) $\rightarrow$ collapsible deliberation log (“How was this plan made?”).

End-of-attempt panel. Replaces colouring UI: final graph render, per-participant score table, shared total, score history across attempts (from attempt 2 onwards), and NEXT ATTEMPT / FINISH SESSION buttons.

6 Data Collection

All events are written to `results/participants/<pid>_<ts>/events.jsonl`. Each line is `{"ts": <float>, "event": "<type>", ...}`. Key events:

Event	Key fields
<code>colour_chosen</code>	node, colour, source , <code>decision_time_s</code> , <code>agents_agreed</code>
<code>plan_modified</code>	node, plan_colour, <code>chosen_colour</code>
<code>explanation_click</code>	node, agent
<code>deliberation_log_opened</code>	attempt
<code>attempt_complete</code>	assignment, score.total , <code>elapsed_s</code>
<code>session_end</code>	score_trajectory , <code>total_attempts</code>

`source` $\in \{\text{human, accepted_plan, modified_plan, agent_auto}\}$. `agents_agreed` is true when a majority of agents recommended the chosen colour. Decision time is measured from `node_ready` to button click.

7 Measures and Hypotheses

Primary: shared score $\eta = \text{Score}_{\text{shared}}/8|V|$.

Secondary: plan-modification rate (% of human steps overridden, Mode 2 only); explanation engagement rate; mean decision latency; $\Delta_k = \text{Score}^{(k)} - \text{Score}^{(1)}$ (improvement across attempts); agents-agreed rate.

Key hypotheses:

- H1.** Modification rate is higher in Mode 2B than 2A (appropriate trust calibration).
- H2.** Mode 2A shared score $>$ Mode 2B; gap grows with better calibration.
- H3.** Mode 1 yields better per-decision calibration than Mode 2 (per-node transparency benefit).
- H4.** High explanation-engagers show better modification calibration in Mode 2.
- H5.** Shared score improves across attempts; improvement larger in Mode 2B.
- H6.** Higher agents-agreed rate correlates with higher shared score but lower individual score.

8 Graph Topologies

Eighteen graph presets span two families and three cluster sizes. *Standard* graphs have dense intra-cluster edges and sparse inter-cluster links (H–A and H–B cross-edges; optional A–B direct link). *Hub* graphs use a star topology per cluster

(one hub connected to all satellites), concentrating constraint propagation through the hub.

Family	n	n_c	Key feature
Standard (s9)	9	3	internal triangles
Standard (s12)	12	4	quad + diagonal
Standard (s18)	18	6	6-cycle + chords
Hub (hub12)	12	4	star + H–A, H–B links
Hub (hub12_tri)	12	4	+ A–B (hub triangle)
Hub (hub15_deep)	15	5	deep cross-constraints
Hub (hub18)	18	6	pairs + full hub triangle
Hub (hub18b)	18	6	asymmetric cluster structure

Each family also has 4-player variants (adds a_3 cluster) and $\times_{ab}$ variants (explicit A–B cross-links reducing human structural centrality).